

Multiple Linear Regression

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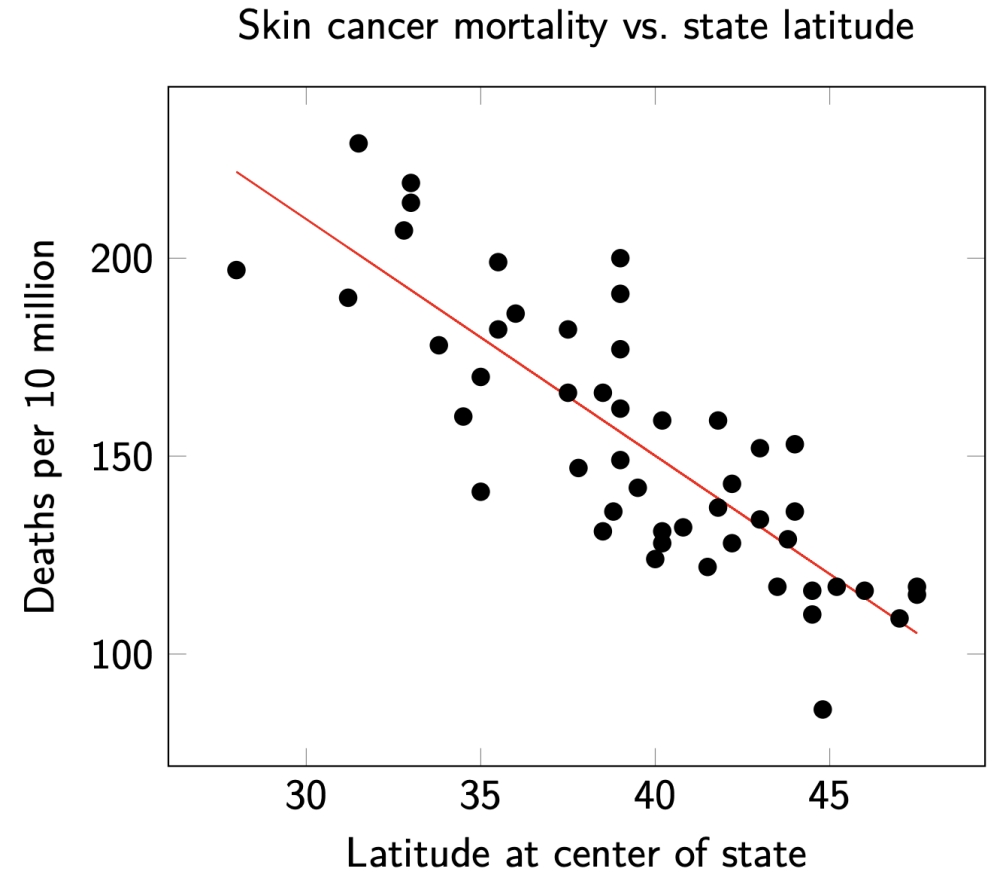
- Recap
- Multiple Linear Regression
- Interpretation of Regression
- Interaction Terms

➤ The slides are based on https://evol.bio.lmu.de/_statgen/Rcourse/ws1617/

Recap Linear Regression

- **Simple regression** considers the relation between a single explanatory variable and response variable

$$X \rightarrow Y$$
$$y = \alpha x + \beta$$



Solving linear regressions

- You learned how to solve linear regressions using the least squares approach
- Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ be the prediction of Y for the i th observation.
- Then the **residual** is defined as

$$e_i = y_i - \hat{y}_i$$

- We define the **residual sum of squares** as

$$\begin{aligned} \text{RSS} &= e_1^2 + e_2^2 + \cdots + e_n^2 = \\ &= (y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_1)^2 + (y_2 - \hat{\beta}_0 - \hat{\beta}_1 x_2)^2 + \cdots + (y_n - \hat{\beta}_0 - \hat{\beta}_1 x_n)^2 \end{aligned}$$

- Note the slightly different notation of β_0 and β_1 instead of β and α .

Solving the least squares regression

- With simple derivation of the RSS formula, we can show that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- with the sample means $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Extensions of the linear model

- You also have looked at some extensions of the linear models:

- Fractional:

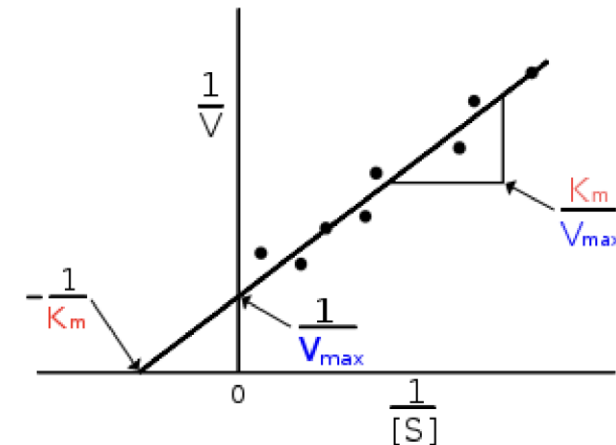
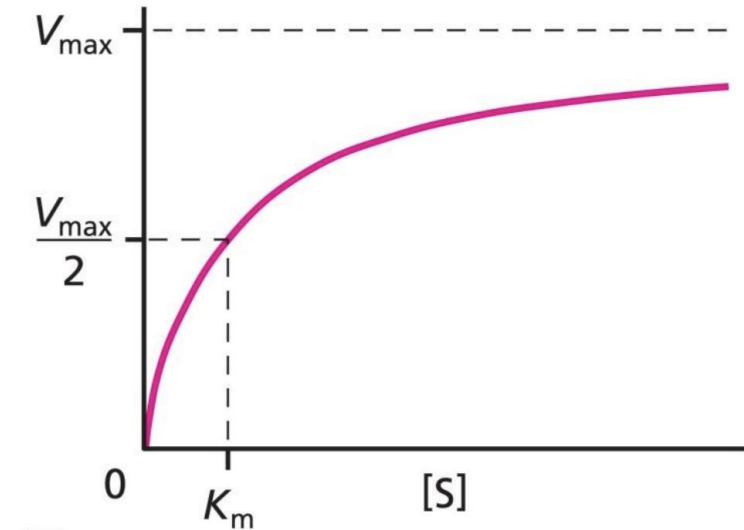
$$y = \frac{ax}{x + b}$$

- Exponential:

$$y = ca^x$$

- Allometric:

$$y = ax^b$$



Limitation of the procedure so far

- With the extensions the problem approach is quite powerful

BUT:

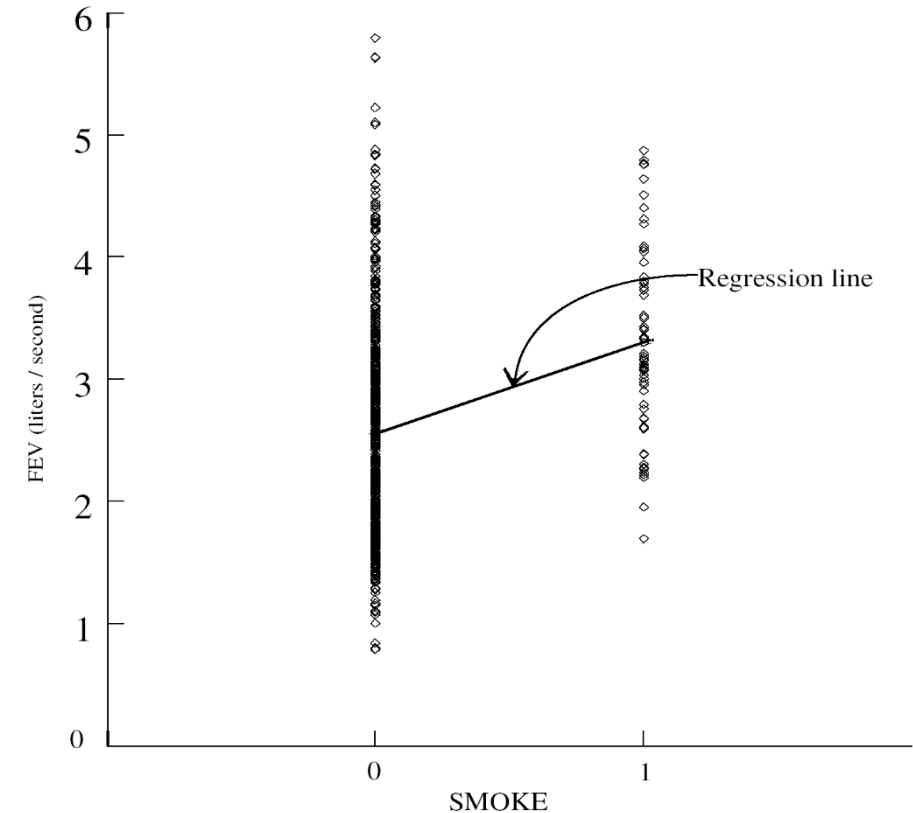
- You only investigate the influence of one variable
- Rarely any problem is governed by only one variable
- The interpretation can even be highly misleading when ignoring other variables

Example

Childhood respiratory health survey:

- Binary explanatory variable (SMOKE) is coded 0 for non-smoker and 1 for smoker
- Response variable Forced Expiratory Volume (FEV) is measured in liters/second
 - The larger the value, the better
- When performing a linear regression on FEV on SMOKE:

$$\hat{y} = 2.634 + 0.605 \cdot S$$



- Example taken from B. Gerstman: https://pdp.sjsu.edu/faculty/gerstman/biostat-text/Gerstman_PP15.ppt

What does this mean?

$$\hat{y} = 2.634 + 0.605 \cdot S$$

- This indicates that smoking positive influences the FEV value
- **Shall we now recommend smoking for children???**
- This is of course utter non-sense:
 - **Children who are smoking tend to be older!**
 - **We have to include the age in the regression analysis**

=> This is where the multiple linear regression comes into play!

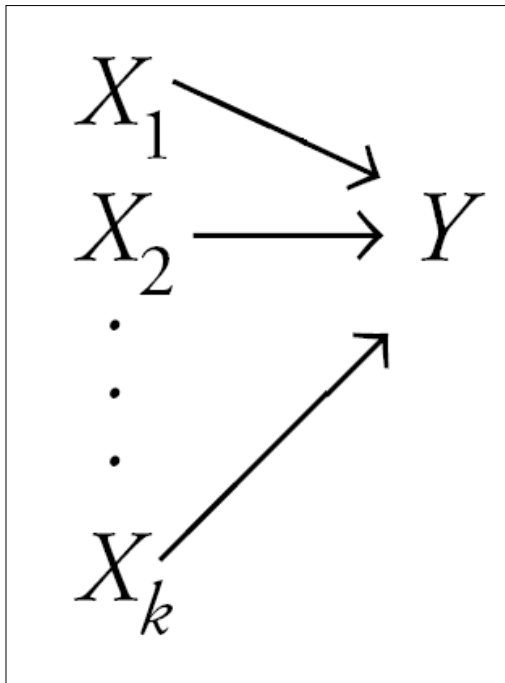
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Say Hello to Multiple Linear Regression



- **Multiple regression** simultaneously considers the influence of multiple explanatory variables on a response variable Y
- The intent is to look at the independent effect of each variable while “adjusting out” the influence of potential confounders
- The method is very similar with what we have seen so far:

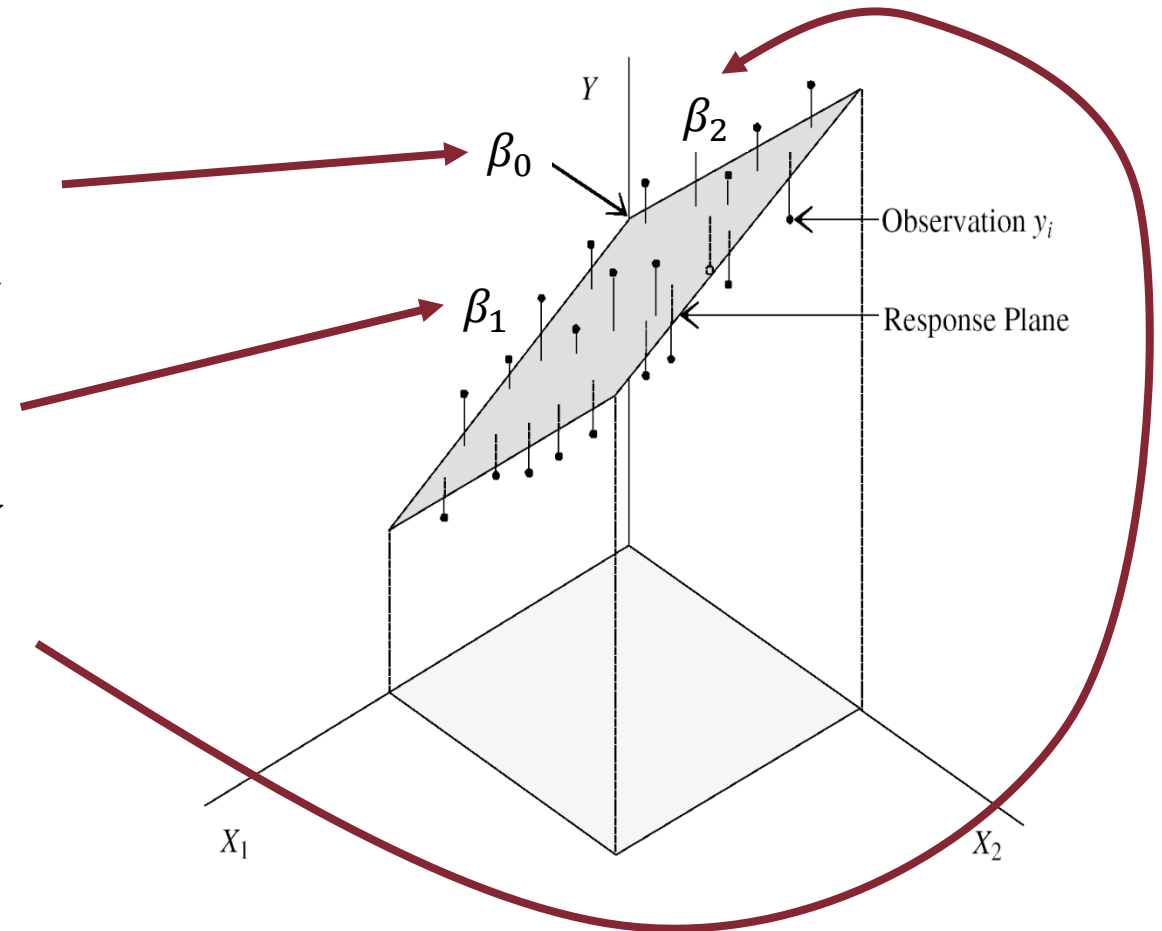
$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$$

- Uses also least squared error for solving

➤ Example taken from B. Gerstman: https://pdp.sjsu.edu/faculty/gerstman/biostat-text/Gerstman_PP15.ppt

Say Hello to Multiple Linear Regression

- Intercept β_0 predicts where the regression plane crosses the Y axis
- The slope for variable X_1 (β_1) predicts the change in Y per unit X_1 holding X_2 constant
- The slope for variable X_2 (β_2) predicts the change in Y per unit X_2 holding X_1 constant
- A multiple regression model with k independent variables fits a regression “surface” in $k + 1$ dimensional space (cannot be visualized)



➤ Example taken from B. Gerstman: https://pdp.sjsu.edu/faculty/gerstman/biostat-text/Gerstman_PP15.ppt

Correcting for age

$$FEV = 0.204 - 0.236 \cdot S + 0.248 \cdot Age$$

- When we include the additional variable into the equation, the things make more sense
 - The slope coefficient associated for SMOKE is -0.236 , suggesting that smokers have $.236$ less FEV on average compared to non-smokers (after adjusting for age)
 - The slope coefficient for AGE is $.248$, suggesting that each year of age is associated with an increase of $.248$ FEV units on average (after adjusting for SMOKE)

=> We will now learn how to work with and interpret multiple linear regressions

- Example taken from B. Gerstman: https://pdp.sjsu.edu/faculty/gerstman/biostat-text/Gerstman_PP15.ppt

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Another Example

- The following examples are taken from the "Introduction to statistical learning" book:

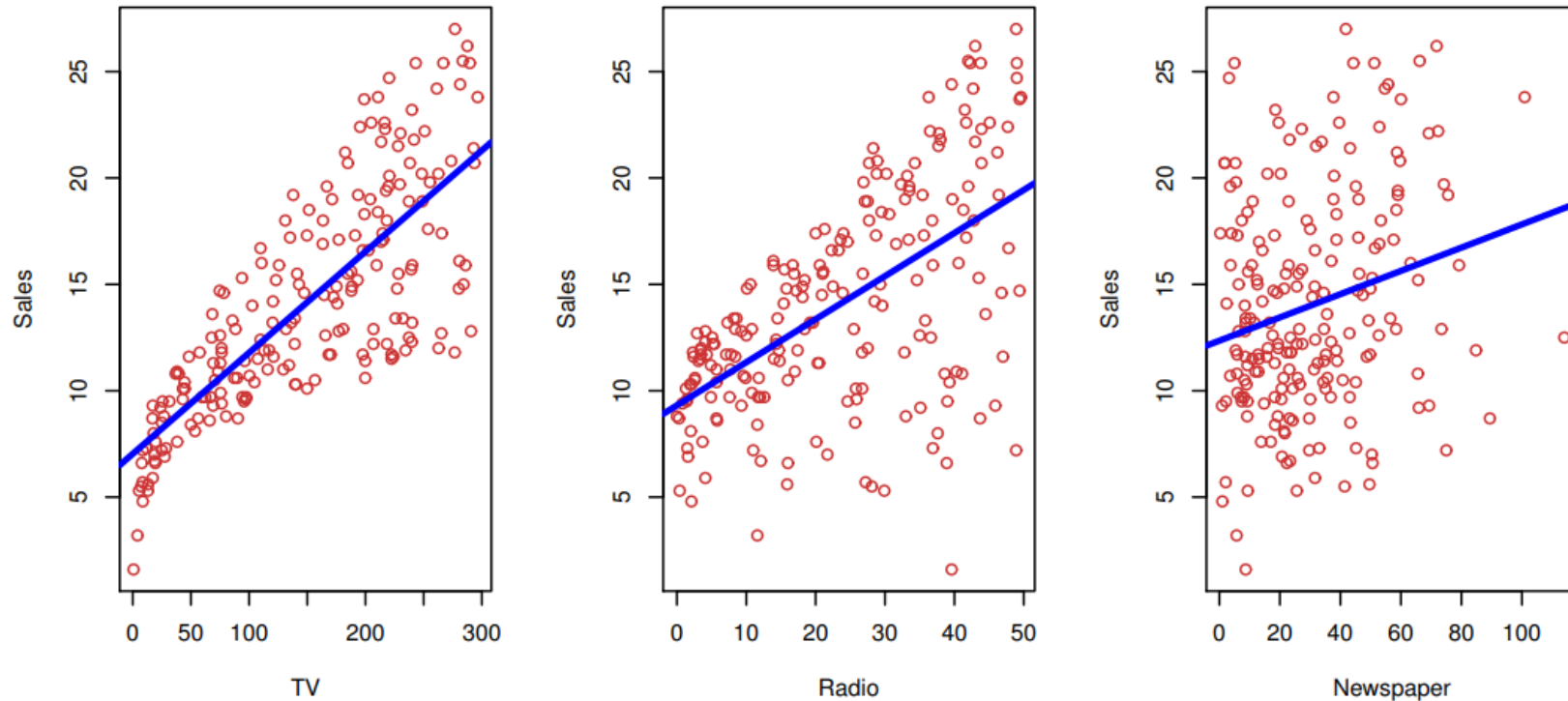
James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning (Vol. 112, p. 18). New York: springer.

- The book is available online for free:

<https://www.statlearning.com/>

- For of particular interest (the entire book is great, but hey): Chapters 3.1 & 3.2

Advertisement Dataset

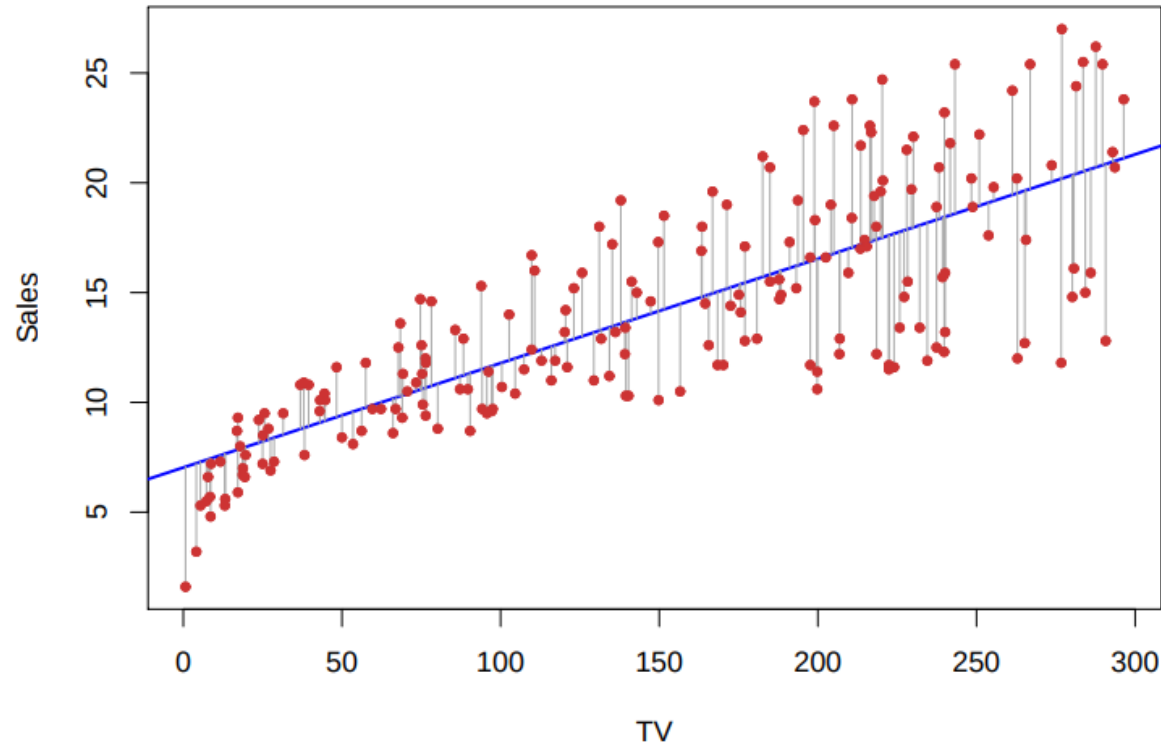


- The Advertising data set consists of the sales of that product in 200 different markets, along with advertising budgets for the product in each of those markets for three different media: TV, radio, and newspaper.

Relevant Questions to ask

- Is there a relationship between advertising budget and sales?
- How strong is the relationship between advertising budget and sales?
- Which media contribute to sales?
- How accurately can we predict future sales?
- Is the relationship linear?
- Is there synergy among the advertising media?

Before we start



- We will first look at the interpretation of the simple linear regression before we move to the multiple regression.

Using R for Linear Regression

```
> model.tv <- lm(Sales ~ TV, data=adv)
> summary(model.tv)
```

```
Call:
lm(formula = Sales ~ TV, data = adv)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.3860	-1.9545	-0.1913	2.0671	7.2124

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.032594	0.457843	15.36	<2e-16	***
TV	0.047537	0.002691	17.67	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.259 on 198 degrees of freedom
Multiple R-squared: 0.6119, Adjusted R-squared: 0.6099
F-statistic: 312.1 on 1 and 198 DF, p-value: < 2.2e-16

➤ Okay, we see that there is a relationship ... maybe

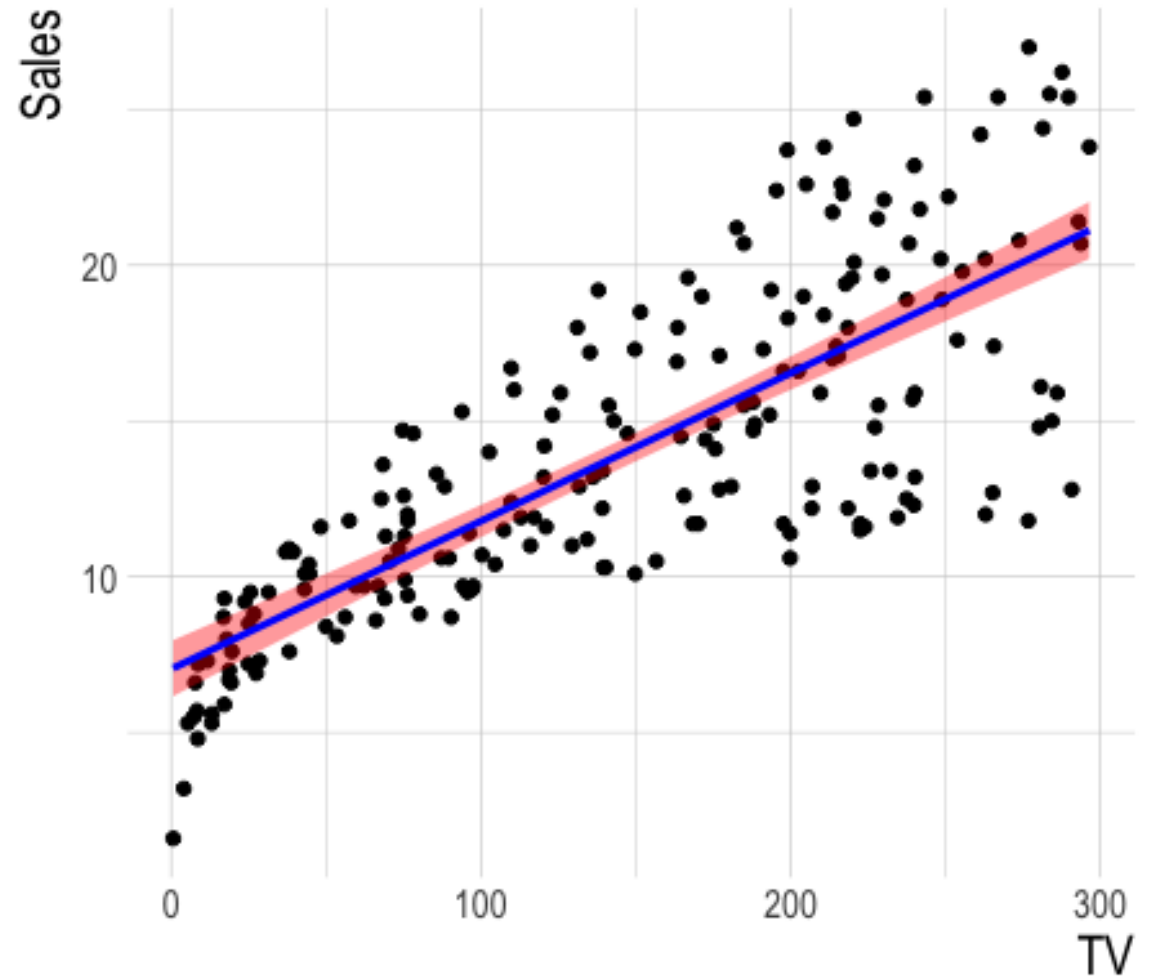
➤ What does all the Information mean?

Accuracy of the Coefficient Estimates

- The standard error of an estimator reflects how it varies under repeated sampling.
- For our example, we have:

$$SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

$$SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$



Confidence Intervals

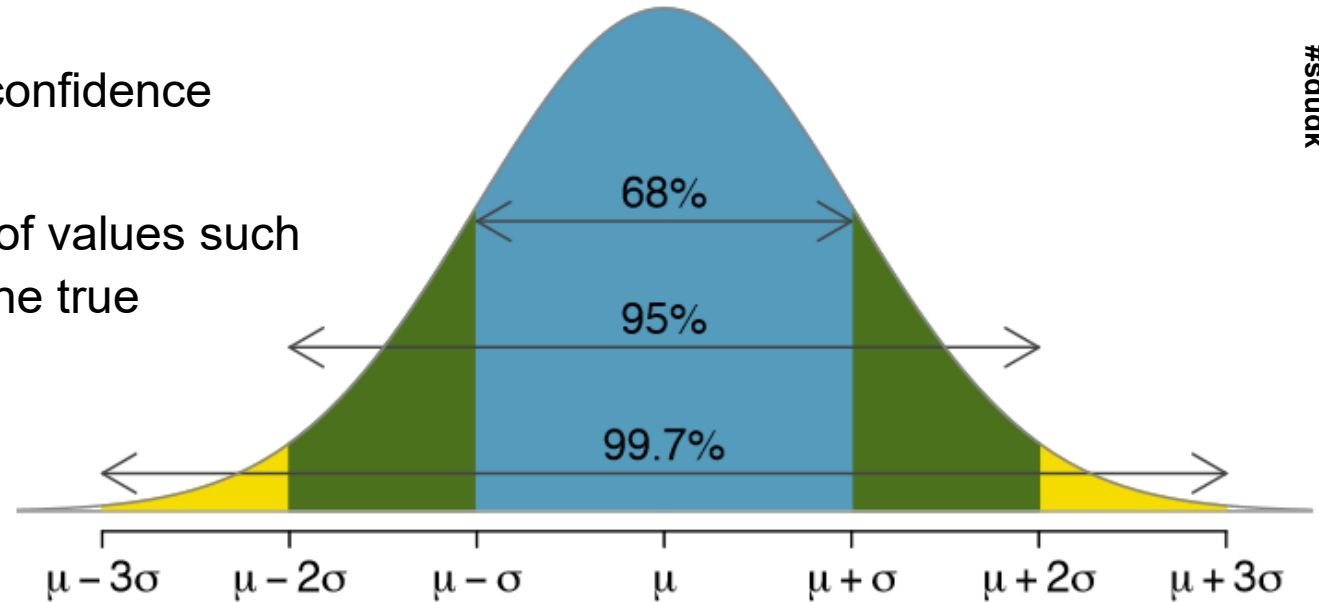
- These standard errors can be used to compute confidence intervals.
- A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter.
- It has the form

$$\hat{\beta}_1 \pm 2 \cdot \text{SE}(\hat{\beta}_1)$$

- In other words: there is approximately a 95% chance that the interval

$$[\hat{\beta}_1 - 2 \cdot \text{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \text{SE}(\hat{\beta}_1)]$$

will contain the true value of $\hat{\beta}_1$.



Hypothesis testing

- Standard errors can also be used to perform hypothesis tests on the coefficients.
- The most common hypothesis test involves testing the **null hypothesis** of:

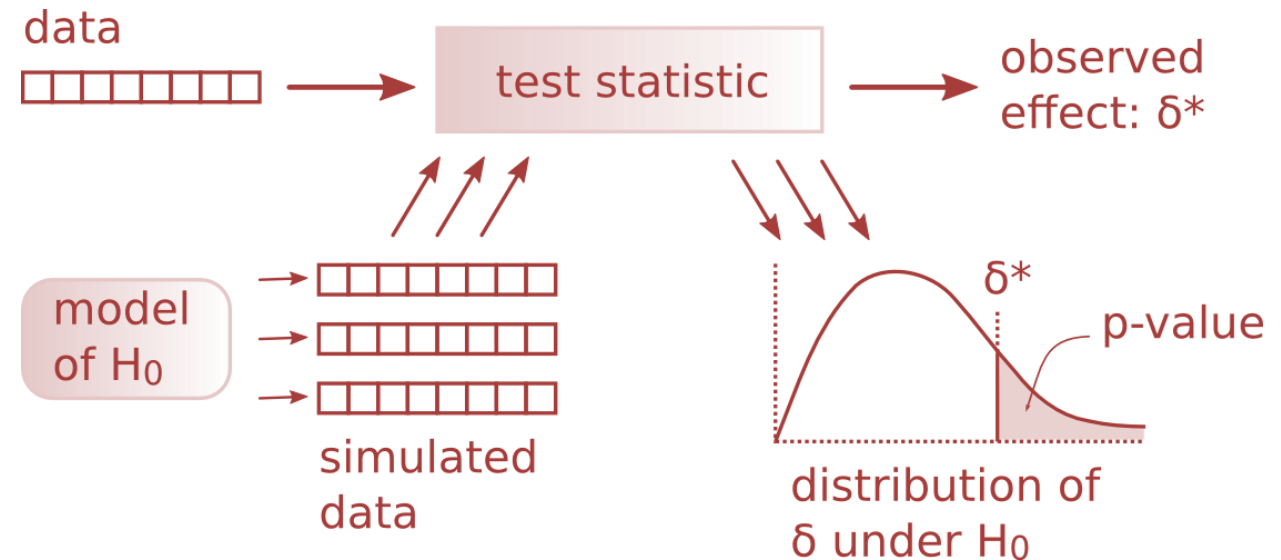
H_0 : There is no relationship between X and Y

$$H_0: \beta_1 = 0$$

versus the alternative hypothesis

H_A : There is some relationship between X and Y

$$H_0: \beta_1 \neq 0$$

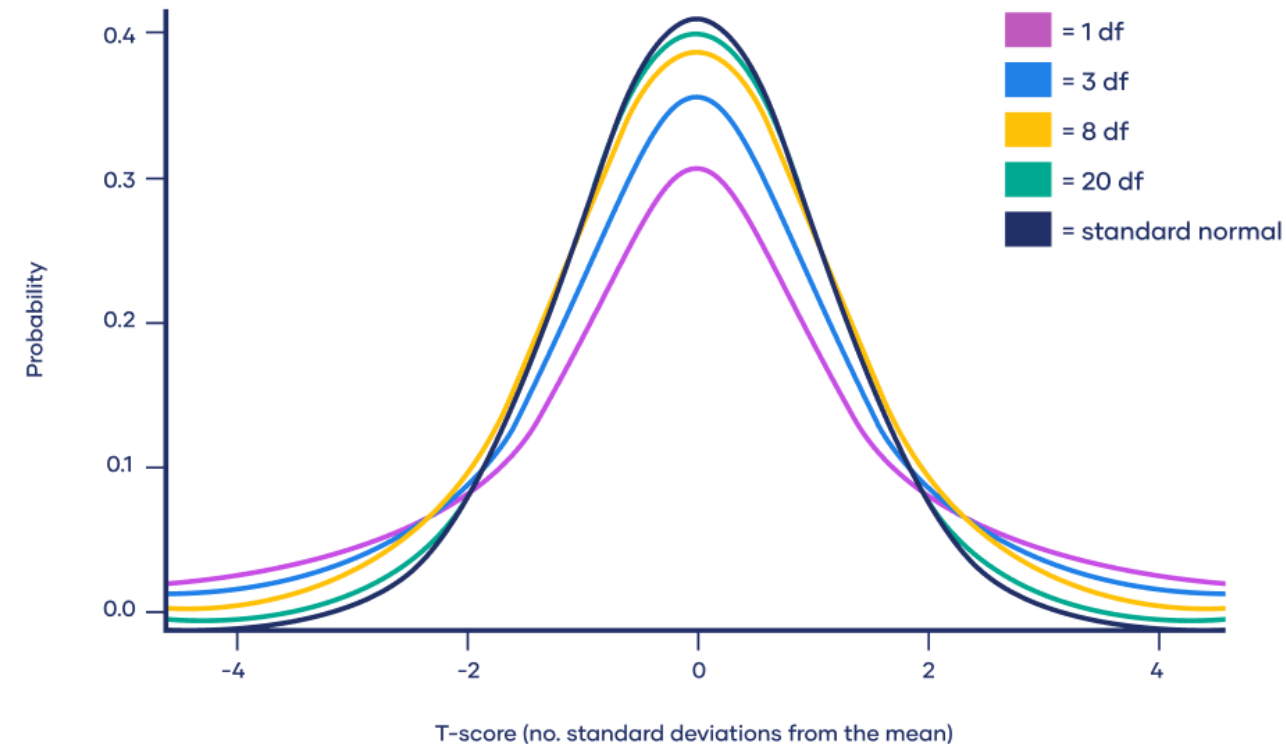


Hypothesis testing

- To test the null hypothesis, we compute a **t-statistic**, given by

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

- This will have a t -distribution with $n - p - 1 = n - 2$ degrees of freedom, assuming $\beta_1 = 0$.
- Using statistical software, it is easy to compute the probability of observing any value equal to $|t|$ or larger. We call this probability the **p-value**.



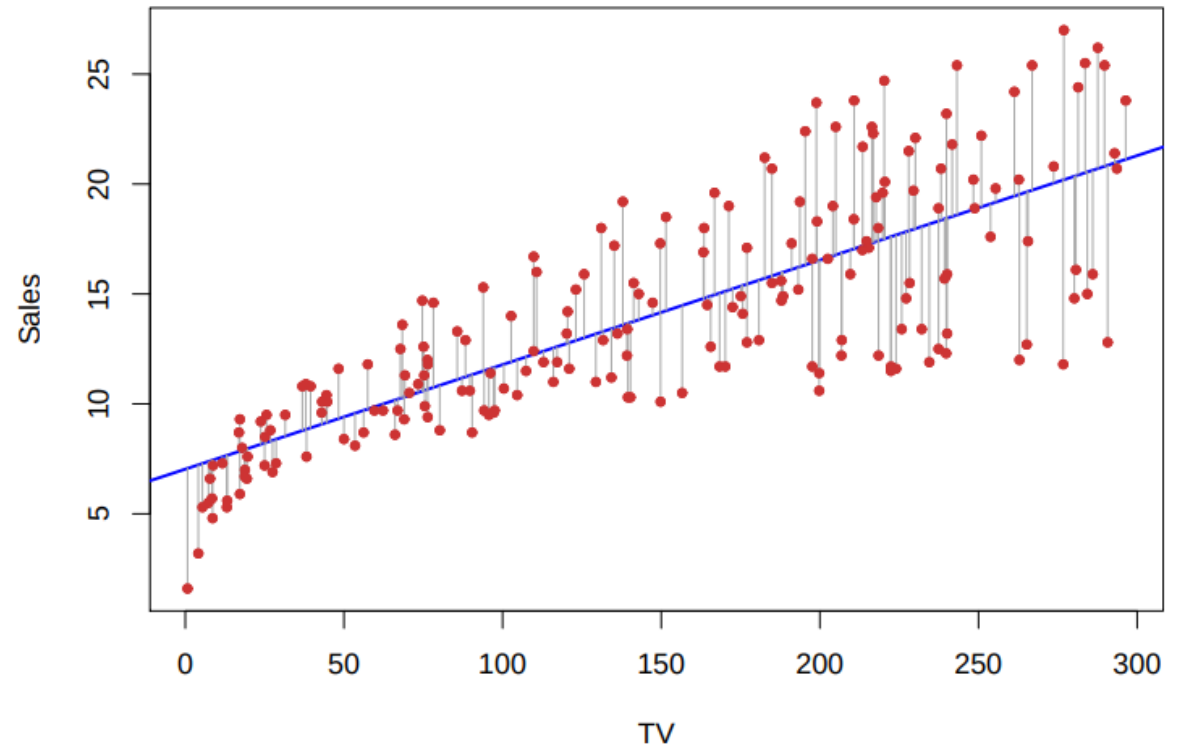
Assessing the Overall Accuracy

- We can compute the **Residual Sum of Squares** (RSS)

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- With that, we can define the **Residual Standard Error** (RSE)

$$\text{RSE} = \sqrt{\frac{1}{n-2} \text{RSS}}$$



Assessing the Overall Accuracy

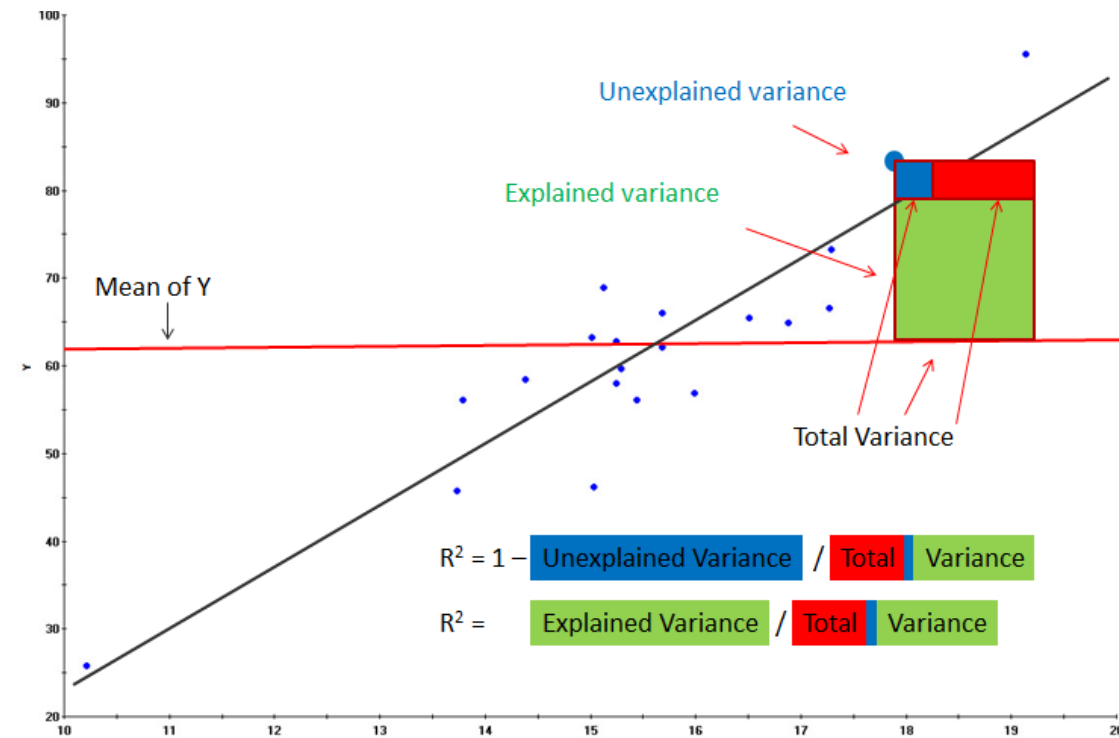
- To assess how much of the variance is explained, we use the **R-squared** measure

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

where $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ is the **total sum of squares**.

- It can be shown that in this simple linear regression setting that $R^2 = r^2$ with r being the **correlation** between X and Y :

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$



Multiple Linear Regression

- We can include several features or predictors into our model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

- We interpret β_j as the average effect on Y of a one unit increase in X_j , holding all other predictors fixed. In the advertising example, the model becomes

$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot \text{newspaper} + \epsilon$$

Multiple Linear Regression in R

```
> model.all <- lm(Sales ~ TV + Radio + Newspaper, data=adv)
> summary(model.all)
```

Call:

```
lm(formula = Sales ~ TV + Radio + Newspaper, data = adv)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.8277	-0.8908	0.2418	1.1893	2.8292

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.938889	0.311908	9.422	<2e-16	***
TV	0.045765	0.001395	32.809	<2e-16	***
Radio	0.188530	0.008611	21.893	<2e-16	***
Newspaper	-0.001037	0.005871	-0.177	0.86	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.686 on 196 degrees of freedom

Multiple R-squared: 0.8972, Adjusted R-squared: 0.8956

F-statistic: 570.3 on 3 and 196 DF, p-value: < 2.2e-16

Results for advertising data

	Coefficient	Std. Error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	-0.001	0.0059	-0.18	0.8599

	Correlations:			
	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

Interpreting Regression Coefficients

“Data Analysis and Regression” Mosteller and Tukey 1977

- A regression coefficient β_j estimates the expected change in Y per unit change in X_j , with all other predictors held fixed. But predictors usually change together!
- Example: Y total amount of change in your pocket; X_1 = # of coins; X_2 = # of pennies, nickels and dimes. By itself, regression coefficient of Y on X_2 will be > 0 . But how about with X_1 in model?
- Y = number of tackles by a football player in a season; W and H are his weight and height. Fitted regression model is $\hat{Y} = b_0 + 0.5 \cdot W - 0.1 \cdot H$. How do we interpret $\beta_2 < 0$?

Two quotes by famous Statisticians

“Essentially, all models are wrong, but some are useful”

George Box

”The only way to find out what will happen when a complex system is disturbed is to disturb the system, not merely to observe it passively”

Fred Mosteller and John Tukey, paraphrasing George Box

Some important questions

1. Is at least one of the predictors $X_1, X_2, \dots, X_p$ useful in predicting the response?
2. Do all the predictors help to explain Y , or is only a subset of the predictors useful?
3. How well does the model fit the data?
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Is at least one predictor useful?

- For the first question, we can use the F-statistic

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)} \sim F_{p, n-p-1}$$

- It's similar to a T statistic from a T-Test; The T-test will tell you if a single variable is statistically significant and an F-test will tell you if a group of variables are jointly significant.
- The F-test is used as a general indicator of the probability that any of the predictor variables contribute to the variance in the dependent variable within the population.
- The null hypothesis is that the predictors' weights are all effectively equal to zero.
- Implying that, none of the predictors contribute to the variance in the dependent variable in the population

About Co-Linearity of Variables

- Highly correlated predictors can lead to collinearity issues, and this can greatly increase the model variance, especially in the context of regression.
- Having correlated variables will result in unnecessarily complex models with more than necessary predictor variables.
 - From a data collection point of view, spending time and money for collecting correlated variables could be a waste of effort.
 - It creates unstable models where statistical inference becomes difficult or unreliable.

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Extensions of the Linear Model

Removing the additive assumption: **interactions** and **nonlinearity**

Interactions:

- In our previous analysis of the Advertising data, we assumed that the effect on sales of increasing one advertising medium is independent of the amount spent on the other media.
- For example, the linear model

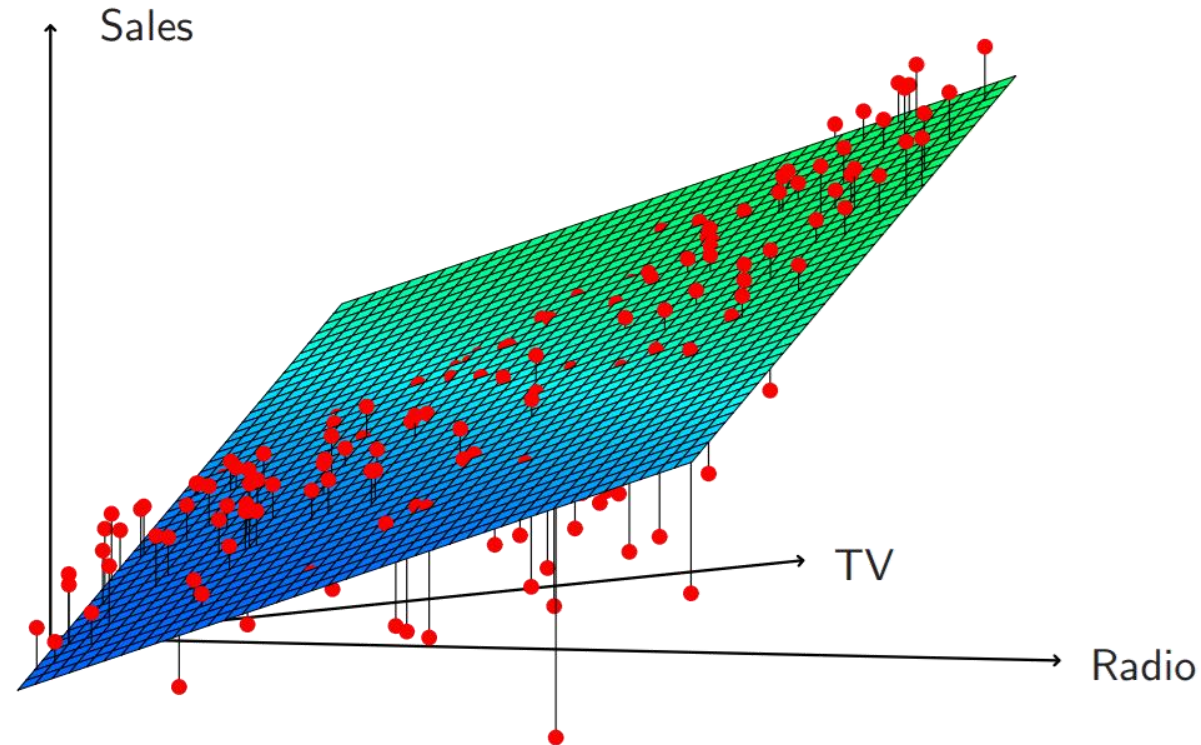
$$\text{sales} = \beta_0 + \beta_1 \cdot \text{TV} + \beta_2 \cdot \text{radio} + \beta_3 \cdot \text{newspaper} + \epsilon$$

states that the average effect on sales of a one-unit increase in TV is always β_1 , regardless of the amount spent on radio.

Interactions

- But suppose that spending money on radio advertising actually increases the effectiveness of TV advertising, so that the slope term for TV should increase as radio increases.
- In this situation, given a fixed budget of \$100,000, spending half on radio and half on TV may increase sales more than allocating the entire amount to either TV or to radio.
- In marketing, this is known as a synergy effect, and in statistics it is referred to as an interaction effect.

Interaction in the Advertising data?



- When levels of either TV or radio are low, then the true sales are lower than predicted by the linear model.
- But when advertising is split between the two media, then the model tends to underestimate sales

New Model

Model takes the form

$$\begin{aligned}\text{sales} &= \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times (\text{radio} \times \text{TV}) + \epsilon \\ &= \beta_0 + (\beta_1 + \beta_3 \times \text{radio}) \times \text{TV} + \beta_2 \times \text{radio} + \epsilon.\end{aligned}$$

Results:

	Coefficient	Std. Error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

Interpretation

- The results in this table suggests that interactions are important.
- The p-value for the interaction term TV×radio is extremely low, indicating that there is strong evidence for $H_A: \beta_3 \neq 0$.
- The R^2 for the interaction model is 96.8%, compared to only 89.7% for the model that predicts sales using TV and radio without an interaction term.

Interpretation — continued

- This means that $(96.8 - 89.7)/(100 - 89.7) = 69\%$ of the variability in sales that remains after fitting the additive model has been explained by the interaction term.
- The coefficient estimates in the table suggest that an increase in TV advertising of \$1, 000 is associated with increased sales of

$$(\beta_1 + \beta_3 \cdot \text{radio}) \times 1000 = 19 + 1.1 \times \text{radio}$$

- An increase in radio advertising of \$1, 000 will be associated with an increase in sales of

$$(\beta_2 + \beta_3 \cdot \text{TV}) \times 1000 = 19 + 1.1 \times \text{TV}$$

Interaction Terms in R

- You can add interactions between variables using “*”
 - This adds the interaction term, as well as the base variables
 - If you only want to add the interaction, use “:”
- You can also use complex modifications to your variables:
 - Some key have special meaning in the formula
 - It is safe to encapsulate those with the I() functions
 - For instance I(TV^2) would use TV in squared fashion

```
> lm(Sales ~ TV*Radio, data=adv)

Coefficients:
(Intercept)          TV          Radio      TV:Radio
   6.750220    0.019101    0.028860    0.001086

> lm(Sales ~ TV:Radio, data=adv)

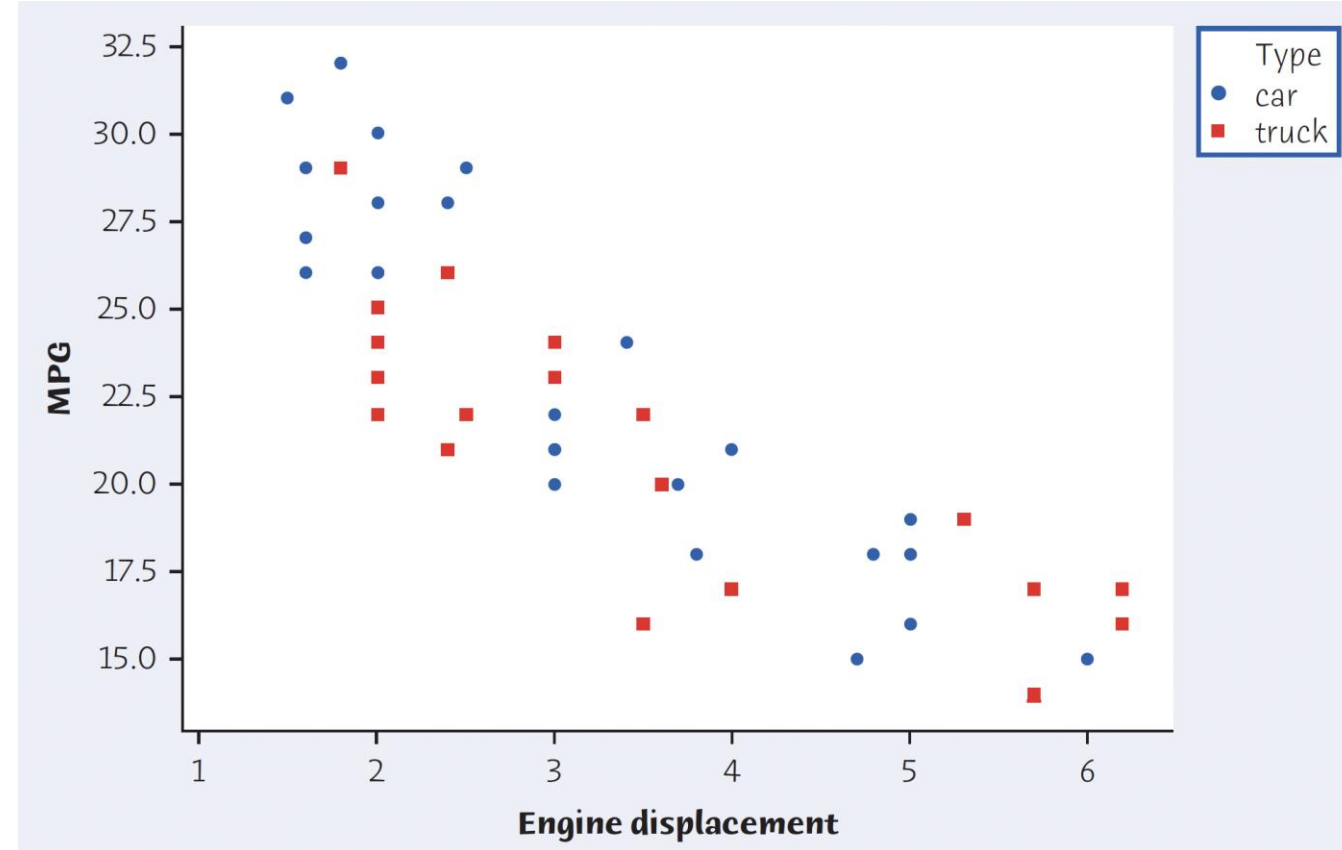
Coefficients:
(Intercept)      TV:Radio
   8.799328    0.001496

> lm(Sales ~ I(TV^2), data=adv)

Coefficients:
(Intercept)      I(TV^2)
   9.6757056    0.0001501
```

Categorical Variables

- Lets look at the “cars” dataset again
- When we want to regress the miles per gallon, the displacement of the engine (basically the size of the engine) is a rather good predictor
- But it makes a difference, whether we are looking at a car or (pick-up)-truck, due to the larger air resistance
- We want to include that in our regression model



Categorical Variables

- A binary variable (category with 2 levels) can be included by encoding

$$x_2 = 0 \text{ for car}$$

$$x_2 = 1 \text{ for truck}$$

- Overall, this leads to the following regression formula (with x_1 being the displacement)

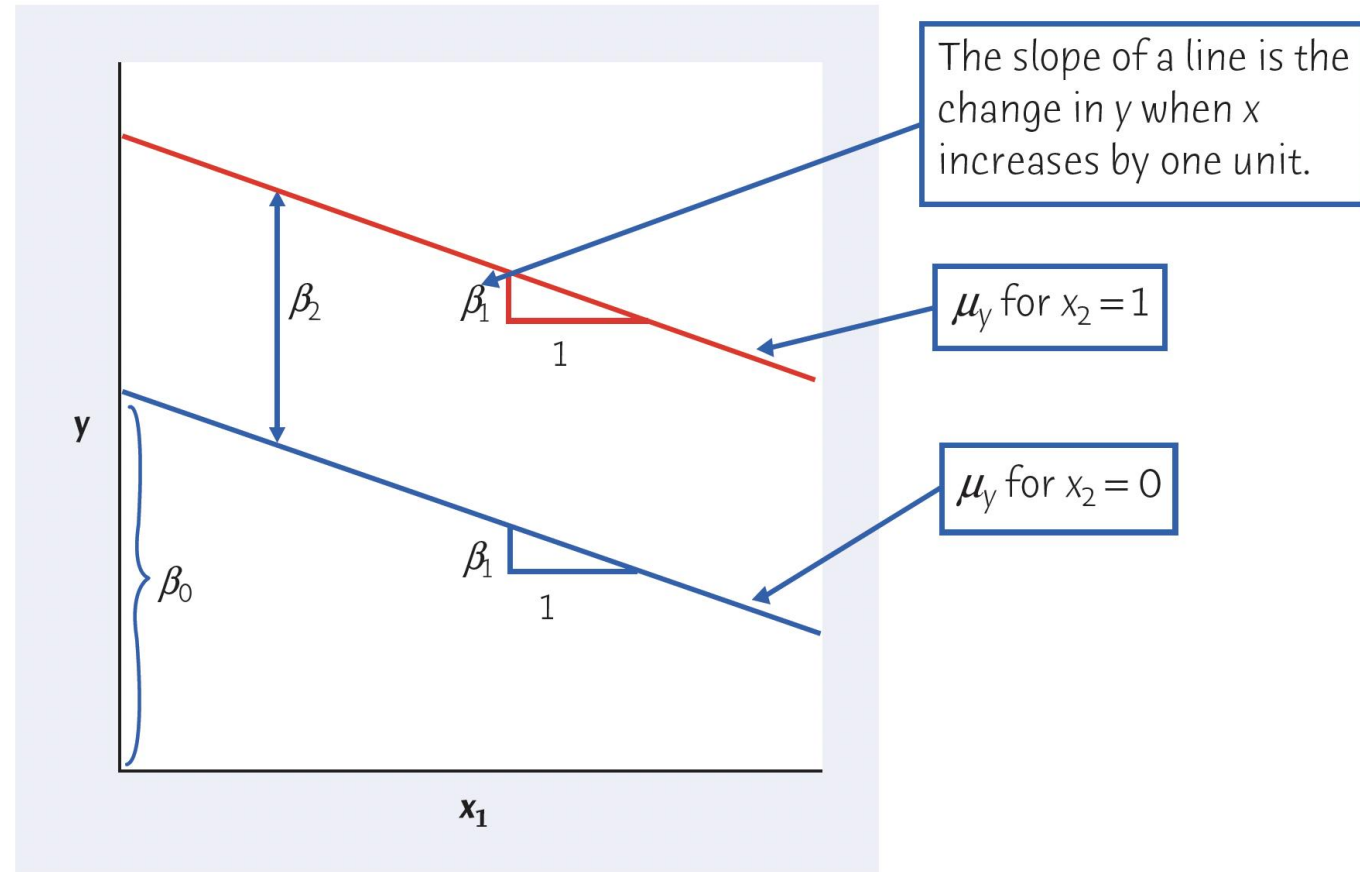
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- For cars ($x_2 = 0$)

$$y = \beta_0 + \beta_1 x_1$$

- For trucks ($x_2 = 1$)

$$y = (\beta_0 + \beta_2) + \beta_1 x_1$$



Categorical Variables

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- For cars ($x_2 = 0$)

$$y = \beta_0 + \beta_1 x_1$$

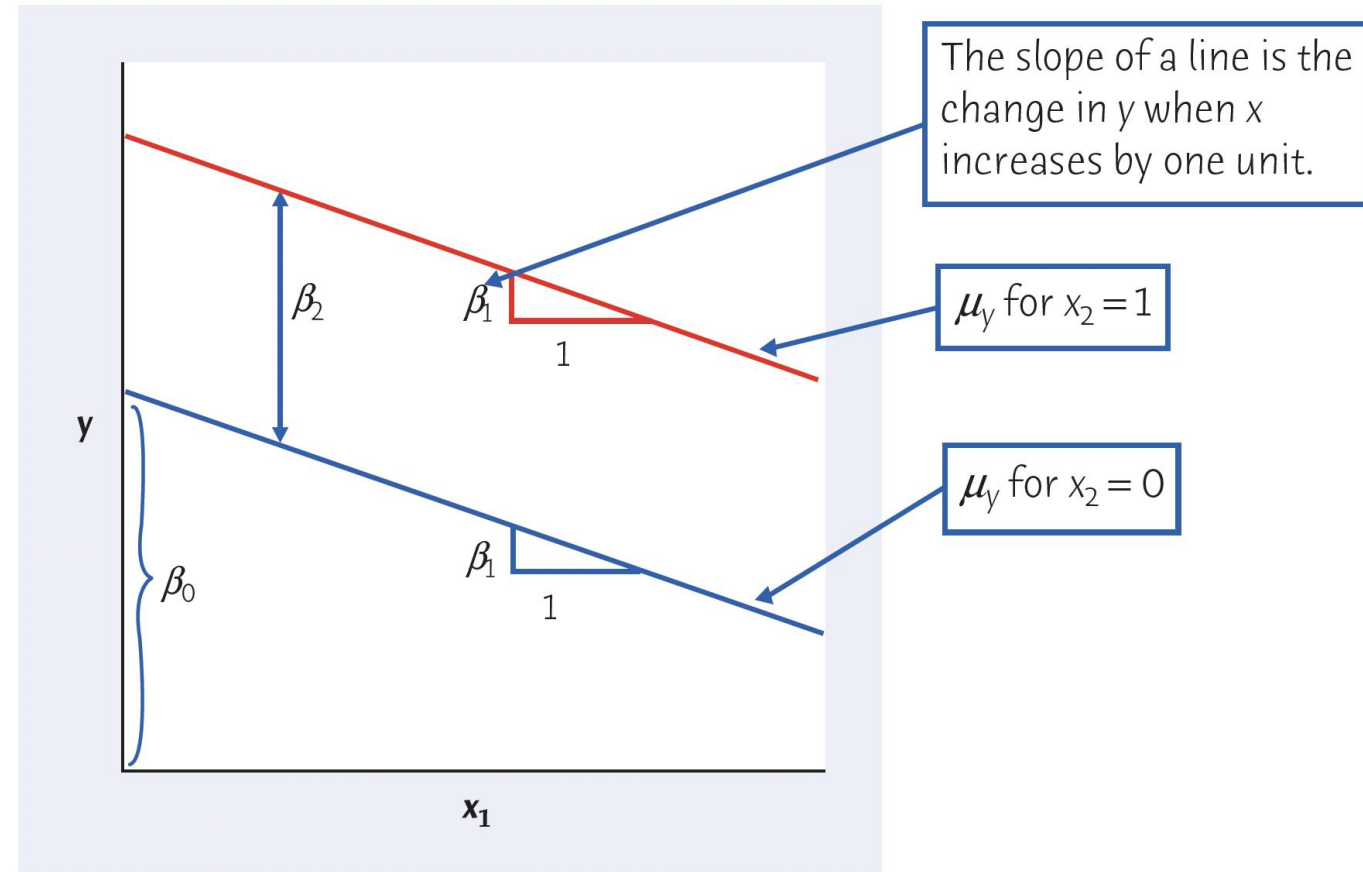
- For trucks ($x_2 = 1$)

$$y = (\beta_0 + \beta_2) + \beta_1 x_1$$

- Consequences:

- We have the same slope for trucks and cars
- We have a different offset for trucks, i.e., trucks are estimated to have a constant offset

- But: Maybe it would be better if trucks would have also a different slope in 'mpg'



Interactions with Categorical Variables

- This can be achieved by including interactions of the categorical variable x_2 and mpg x_1 leading to the following model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

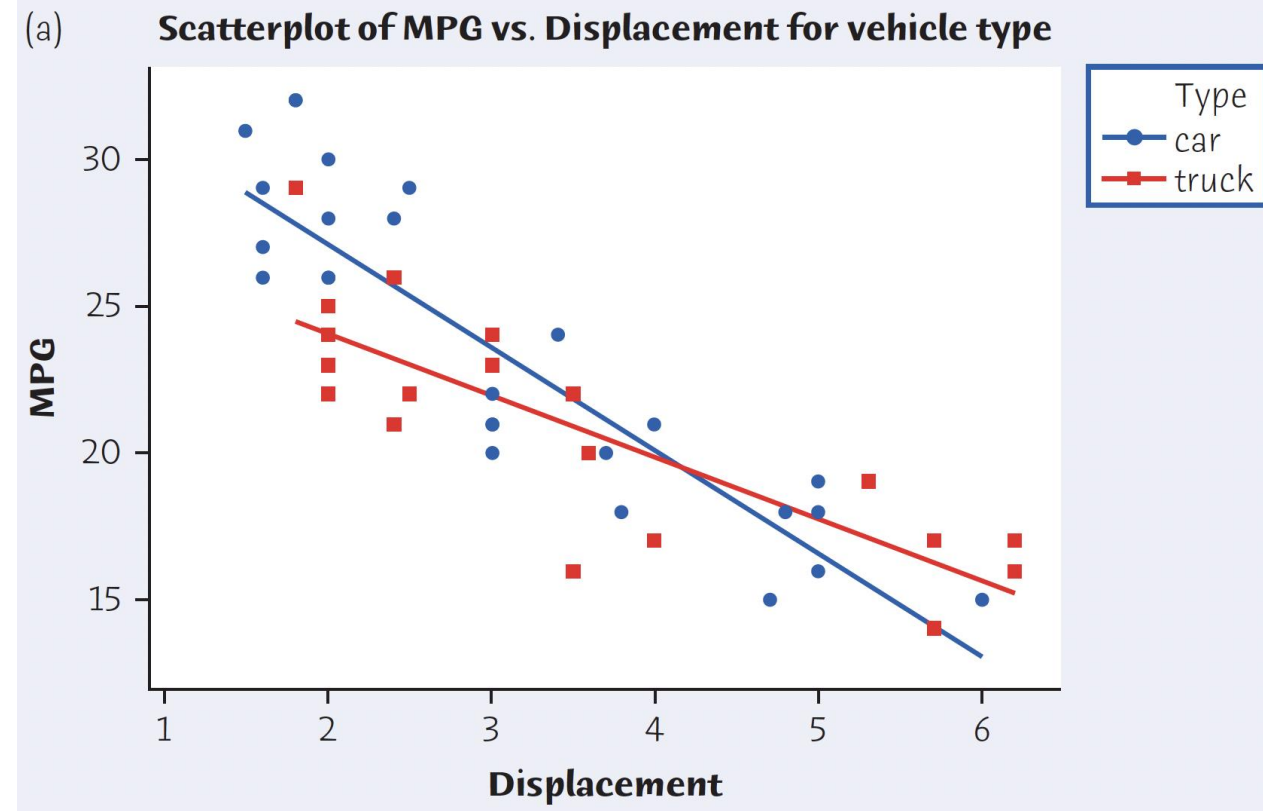
- For cars ($x_2 = 0$)

$$y = \beta_0 + \beta_1 x_1$$

- For trucks ($x_2 = 1$)

$$y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x_1$$

- With that, we have a modification of the slope by β_3



Categorical variables with more levels

- For categorical variables with more levels, for instance the origin of the cars:

origin = {*Europe*, *Japan*, *US*}

- Here, we create (or rather R creates) dummy variables for you:
 - *originJapan* = {0,1}
 - *originUS* = {0,1}
- Note the absence of “*originEurope*”, since the first level in your factor is the baseline

```
> summary(lm(mpg ~ displacement + origin, data=auto))

Call:
lm(formula = mpg ~ displacement + origin, data = auto)

Residuals:
    Min       1Q   Median       3Q      Max
-14.2925  -2.8105  -0.6411   2.1874  18.8230

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  34.037407   0.634248  53.666  < 2e-16 ***
displacement -0.056311   0.002914 -19.327  < 2e-16 ***
originJapan   2.196898   0.753881   2.914  0.00377 **
originUS     -0.106829   0.737955  -0.145  0.88497
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.591 on 394 degrees of freedom
Multiple R-squared:  0.6575,    Adjusted R-squared:  0.6549
F-statistic: 252.2 on 3 and 394 DF,  p-value: < 2.2e-16
```

Q&A

contact: roettger@imada.sdu.dk